

# Arithmetic Sequences as Linear Functions

## Answer Key

### Algebra 1 Sequences and Linear Modeling

#### Quick Answers

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|--------|--------|--------|---------------|
| 1. 33  | 4. 11  | 7. 103 | 10. 4         |
| 2. 31  | 5. -13 | 8. 11  | 11. 615       |
| 3. 100 | 6. 348 | 9. 820 | 12. 405, 5250 |

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#### Curriculum

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**Level:** Algebra 1 Sequences and Linear Modeling

HSF-BF.A.2, HSF-LE.A.2 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12*

*Difficulty 1–5 of 5 across 13 tagged checks.*

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#### Common wrong answers

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1. *If they got 38: used  $n$  instead of  $n-1$ , adding one common difference too many*
  5. *If they got 53: added the common difference instead of subtracting it*
  6. *If they got 216: multiplied the first row by the number of rows and ignored the increase*
  9. *If they got 1580: multiplied the last term by the number of terms instead of using the average of first and last*
  11. *If they got 930: used 15 rows of the last row's 62 seats instead of averaging first and last*
  12. *If they got 8100: multiplied the week-20 output by 20 instead of summing the twenty different weekly outputs*
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#### 1. 7th term of the sequence with $a_1 = 3$ , $d = 5$ .

**Step 1.** The question asks for one specific term, so the explicit formula  $a_n = a_1 + (n - 1)d$  is the right tool; identify  $a_1 = 3$ ,  $d = 5$ ,  $n = 7$  before substituting.

**Step 2.**  $a_7 = 3 + (7 - 1)(5) = 3 + 30 = 33$ .

33

*Why  $n - 1$ :* getting from term 1 to term 7 takes six steps, not seven. Using  $n$  instead gives 38, one common difference too many.

#### 2. $f(10)$ for $f(n) = 3n + 1$ .

**Step 1.** The sequence has already been packaged as a linear function of the term number, so the work is a single substitution rather than a recursion.

**Step 2.**  $f(10) = 3(10) + 1 = 30 + 1 = 31$ .

31

*Reading the form:* the slope 3 is the common difference, and the constant 1 is the value the line would take at  $n = 0$  — one step *before* the first term, which is why it is not the first term itself.

**3.**  $1 + 3 + 5 + \cdots + 19$ .

**Step 1.** This is a total, not a term, so pair the ends: every pair of terms equally far from the two ends has the same sum, which turns the addition into a multiplication.

**Step 2.**  $1 + 19 = 20$ ,  $3 + 17 = 20$ ,  $5 + 15 = 20$ , and so on. With 10 terms there are 5 such pairs, each worth 20.

**Step 3.**  $5 \times 20 = 100$ . 100

*General form:*  $S_n = \frac{n(a_1 + a_n)}{2} = \frac{10(1 + 19)}{2} = 100$  — the same pairing written once and for all.

**4. Which term of 5, 9, 13, ... is 45?**

**Step 1.** Here the term *value* is known and the term *number* is not, so write the explicit formula as an equation and solve for  $n$  rather than listing terms.

**Step 2.**  $5 + (n - 1)(4) = 45$ , so  $4n + 1 = 45$ , then  $4n = 44$  and  $n = 11$ . 11

*Sanity check:*  $n$  must be a positive whole number. If it came out as 10.5, the value 45 would simply not be in the sequence.

**5. 12th term of a sequence starting at 20 and decreasing by 3.**

**Step 1.** “Decreases by 3” makes the common difference negative, so record  $d = -3$  before substituting — the formula itself does not change.

**Step 2.**  $a_{12} = 20 + (12 - 1)(-3) = 20 - 33 = -13$ . -13

*The common error:* using  $d = +3$  gives 53, and a sequence that was described as decreasing would have grown. Check the sign against the story.

**6. Auditorium: 12 rows, 18 seats in row 1, 2 more per row.**

**Step 1.** Part (b) asks for a total across all rows, so this is a sum; but the sum formula needs the last term, so find that first.

**Step 2.** Last row:  $a_{12} = 18 + (12 - 1)(2) = 18 + 22 = 40$  seats.

**Step 3.** Total:  $S_{12} = \frac{12(18 + 40)}{2} = 6 \times 58 = 348$ . 348

*The trap:*  $12 \times 18 = 216$  counts every row as the front row and throws away the growth entirely.

**7. 7, 11, 15, ... as  $f(n) = 4n + 3$ ; find  $f(25)$ .**

**Step 1.** Identify the two constants from the sequence itself: the terms rise by 4 each step, so the slope is 4, and the constant is chosen so that  $f(1)$  returns the first term.

**Step 2.**  $f(1) = 4(1) + 3 = 7$ , which confirms the form. Then  $f(25) = 4(25) + 3 = 100 + 3 = 103$ . 103

*Where the numbers come from:* the 4 is the common difference (the slope), and the 3 is  $a_1 - d = 7 - 4$ , the imaginary “term zero”.

**8. Priya’s savings reach 500 dollars in which month?**

**Step 1.** The balance is arithmetic with  $a_1 = 150$  and  $d = 35$ , and the unknown is the month number, so solve the explicit formula for  $n$ .

**Step 2.**  $150 + (n - 1)(35) = 500$ , so  $35n + 115 = 500$ , then  $35n = 385$  and  $n = 11$ . 11

*Check:*  $n = 11$  is a whole number, so the balance hits 500 exactly at the end of month 11. Had  $n$  come out fractional, the answer would be the next whole month, since deposits arrive in whole steps.

**9. Sum of the first 20 terms of 3, 7, 11, ...**

**Step 1.** The sum formula needs the last term, so compute  $a_{20}$  before summing anything.

**Step 2.**  $a_{20} = 3 + (20 - 1)(4) = 3 + 76 = 79$ .

**Step 3.**  $S_{20} = \frac{20(3 + 79)}{2} = 10 \times 82 = 820$ .

820

*The trap:*  $20 \times 79 = 1580$  treats every term as if it were the largest one. The correct factor is the *average* of the first and last terms, 41, not the last term itself.

**10. 4th term 14, 9th term 34 — find  $d$  and  $a_1$ .**

**Step 1.** Neither the first term nor the common difference is given, but the gap between two known terms is made only of common differences, so count the steps rather than guessing  $a_1$ .

**Step 2.** From term 4 to term 9 is  $9 - 4 = 5$  steps, so  $14 + 5d = 34$ , giving  $5d = 20$  and  $d = 4$ .

4

*Part (c):* now  $a_4 = a_1 + 3d$ , so  $14 = a_1 + 12$  and the first term is  $a_1 = 2$ . The sequence is 2, 6, 10, 14, ..., and indeed its 9th term is  $2 + 8(4) = 34$ .

**11. Theatre: 15 rows, 20 seats in row 1, 3 more per row.**

**Step 1.** Part (a) asks for a rule, not a number, so write the explicit formula with the row number as the variable: seats in row  $k$  is  $20 + (k - 1)(3)$ , which simplifies to  $3k + 17$ .

**Step 2.** Last row:  $20 + (15 - 1)(3) = 20 + 42 = 62$  seats.

**Step 3.** Capacity:  $S_{15} = \frac{15(20 + 62)}{2} = \frac{15 \times 82}{2} = 615$ .

615

*The trap:*  $15 \times 62 = 930$  fills every row to the size of the back row. The average row holds 41 seats, not 62.

**12. Challenge: factory output, 120 units in week 1, +15 per week.**

**Step 1.** Part (a) asks for a single week's output and part (b) for a running total, so separate the two tools before computing: explicit formula first, sum formula second.

**Step 2.** Week 20:  $a_{20} = 120 + (20 - 1)(15) = 120 + 285 = 405$  units.

405

**Step 3.** Total over 20 weeks:  $S_{20} = \frac{20(120 + 405)}{2} = 10 \times 525 = 5250$  units.

5250

*Part (c):* multiplying the week-20 output by 20 gives 8100, which assumes every week matched the best week. The correct idea is to multiply by the *average* weekly output,  $\frac{120 + 405}{2}$ , which is what the sum formula does.